

CHAPTER 9

The Price Puzzle: What Drives the Market

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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The Big Questions

1 **What are the factors that influence the demand for and supply of goods and services in a market?**

✔ ANSWER

Demand: price of the good, prices of related goods (substitutes and complements), income, tastes and preferences, size and composition of population, seasonality, future price expectations.

Supply: price of the good, prices of related goods, number of sellers, technology, input costs, future expectations and things like weather.

2 **How are prices of goods and services determined through demand and supply interactions?**

✔ ANSWER

Prices come from the interaction of buyers and sellers.

- When supply is less than demand (shortage), prices rise.
- When supply is more than demand (surplus), prices fall.
- The price settles where quantity demanded = quantity supplied (equilibrium price).

3 **What is market equilibrium, and does it exist in the real world?**

✔ ANSWER

Market equilibrium is the point where quantity demanded equals quantity supplied: no shortage, no surplus, no pressure on prices to change.

In the real world markets keep changing (technology, wages, weather, wars, pandemics), so equilibrium is never stable; the market is always adjusting to a new one. Example: face mask prices in COVID-19 (2020).

4 **How and why does the government intervene in the market?**

✔ ANSWER

Markets do not always work fairly, so the government steps in for welfare of vulnerable and low-income groups:

- Regulates unfair practices: price ceiling (e.g. medicines), price floor (minimum wage), control of monopolies.
- Provides public goods (roads, parks, defence, streetlights).
- Regulators like RBI, TRAI, SEBI ensure transparency.

But too much intervention can harm (price distortions, compliance burden, less innovation).

⭐ Don't Miss Out · Demand

? Did you notice that the market demand curve is flatter than Srivalli's individual demand curve? Why is that?

✔ ANSWER

Market demand adds up many consumers, so the same price change gives a bigger change in total quantity.

When price falls from ₹150 to ₹50, Srivalli buys 2 kg more, but market demand rises by 12 kg (6 kg → 18 kg). So the market curve is flatter.

🤔 Think About It · Other Determinants of Demand

? What happens when you consume the first mango? The second one? The third one and so on? You are barely interested in eating mangoes by this point. Why do you think this happens?

✔ ANSWER

The extra satisfaction (utility) from each additional mango keeps falling as we eat more.

This is the **diminishing marginal utility principle**. As utility falls, our willingness to pay also falls, so demand falls.

⭐ Let's Explore · Other Determinants of Demand

1 **Create your own demand schedule for buying notebooks at different prices. At what price would you buy the most? At what price would you stop buying altogether? What could be the reason behind your choices?**

💡 SAMPLE ANSWER

Price of a notebook	Notebooks I would buy
₹10	8
₹20	6
₹40	4
₹60	2
₹100	0

I buy the most (8) at ₹10 and stop buying at ₹100.

Reason: my pocket money is limited (purchasing power); at a high price I would use old notebooks or loose sheets instead. This follows the Law of Demand.

2 **Ask your family members if they postponed or advanced buying any product because of future expectations of changes in price.**

💡 SAMPLE ANSWER

Example: My parents waited to buy a TV until the Diwali sale because they expected discounts (postponed).

Another time they bought extra cooking oil early because they heard its price would rise (advanced).

This shows future price expectations change present demand.

⭐ Let's Explore · Other Determinants of Supply

? **What happens to the supply of a product in case of a change in the cost of inputs, discovery of an alternate input, depletion of resources, change in weather, disaster, etc.? Discuss in class using examples.**

💡 SAMPLE ANSWER

- **Input cost rises** (e.g. fertiliser, diesel): production costs more, supply falls. Input cost falls → supply rises.
- **Cheaper alternate input found** (e.g. solar power for factories): cost falls, supply rises.
- **Depletion of resources** (e.g. groundwater, fish stocks): less can be produced, supply falls.

- **Bad weather** (e.g. heavy rain spoils tomatoes): supply falls, prices rise; good monsoon → more supply.
- **Disaster** (flood, cyclone, pandemic): production and transport stop, supply falls.

⭐ Let's Analyse · Market Equilibrium

? Using data from Table 9.3, plot the demand and supply curve at the three prices, i.e., ₹40, ₹100 and ₹150. Identify and mark excess demand and supply on the graph. Think about how equilibrium could be reached.

Price (₹)	Qd (kg)	Qs (kg)
40	38	6
100	12	12
150	8	43

✔ ANSWER

Price on the y-axis, quantity on the x-axis. Demand points: (38, ₹40), (12, ₹100), (8, ₹150) → downward curve. Supply points: (6, ₹40), (12, ₹100), (43, ₹150) → upward curve.

- They cross at ₹100, 12 kg: **equilibrium**.
- At ₹40: Qd 38 > Qs 6 → **excess demand of 32 kg** (gap between the curves below equilibrium). Buyers compete, so price rises towards ₹100.

- At ₹150: Qs 43 > Qd 8 → **excess supply of 35 kg** (gap above equilibrium). Sellers cut prices to sell, so price falls towards ₹100.

🤔 Think About It · Does Market Equilibrium Exist in the Real World?

1 **Can you think of another real-life example (other than hotels) where prices change frequently? Explain why the prices keep changing.**

💡 SAMPLE ANSWER

Flight tickets (or app cab fares). The same seat can cost ₹3,000 one day and ₹9,000 another.

Prices change with demand: holidays and festivals, how many seats are left, days left before travel, fares of other airlines, fuel cost.

High demand / few seats → higher fare; low demand → discounts.

2 **Our choices today affect future resources (fast fashion, overfishing, overuse of groundwater can harm future supply). Should we focus only on short-term gains, or also think about long-term sustainability? How could this affect the market equilibrium?**

💡 SAMPLE ANSWER

We should also think about long-term sustainability.

If resources are overused now, future supply will fall. Lower supply with the same demand means shortages and higher equilibrium prices later (e.g. costly fish, water scarcity).

Using resources wisely keeps supply steady and prices stable for the future.

🤔 Think About It · Role of Government in the Economy

1 **Have you ever seen or heard of the government fixing prices or wages (for example, bus fares, medicines, or minimum wages)? Share an example and why you think it was done.**

💡 SAMPLE ANSWER

Example: government buses charge fixed fares, and the government fixes the maximum price of essential medicines.

Reason: so that poor and low-income people can afford travel and medicines, and sellers cannot overcharge. Minimum wages make sure workers earn enough for their work.

2 **Do you remember some regulators from the Grade 7 Social Science textbook chapter 'Understanding Markets'?**

💡 SAMPLE ANSWER

Regulators named in this chapter: Reserve Bank of India (RBI) for banking, Central Consumer Protection Authority (consumer rights and unfair trade), TRAI (telecom), SEBI (securities market).

(Also recall others from your Grade 7 book, e.g. FSSAI for food safety.)

⭐ Don't Miss Out · Regulation of Unfair Practices

? During COVID-19 the government capped the price of sanitisers at ₹100 for 200 ml under the Essential Commodities Act, 1955. How do such price controls affect suppliers and consumers? Do you think such controls should be in practice forever?

💡 SAMPLE ANSWER

- **Consumers:** get essential goods at fair prices; hoarding and black-marketing are stopped.
- **Suppliers:** earn less profit per unit; some may supply less, but here many companies started production and sanitisers became widely available.

Such controls are useful in an emergency but should not continue forever: low fixed prices can reduce producers' incentive, cause shortages and stop innovation. Once supply catches up, the market should set prices.

⭐ Let's Explore · Provision of Public Goods

? **From your surroundings, list two goods or services provided by the government. Choose one and answer: Who benefits from it? Why would it be difficult for a private company to provide it on its own? If the government stops providing it, what problems might people face?**

💡 SAMPLE ANSWER

Two goods: **streetlights** and **public park**. Chosen: streetlights.

- **Who benefits:** everyone who walks or drives at night: students, workers, shopkeepers, women, the elderly.
- **Why not a private company:** it cannot charge each person who uses the light; people would use it without paying, so there is no direct profit.
- **If it stops:** dark streets, more accidents, theft and crime, people unsafe going out at night.

⭐ Let's Recall · Limitations of Government Intervention

1 **According to you, how should a democratic government decide when and how much it should intervene in markets to protect people's welfare?**

💡 SAMPLE ANSWER

- Intervene when markets fail or are unfair: essentials becoming unaffordable, monopolies, hoarding, public goods not provided.
- Intervene only as much as needed, often for a limited time (as with sanitisers in COVID-19).
- Use data, consult people, and review the effect so that rules do not hurt producers or small businesses.

2 **Whose voices should a democratic government consider while making such decisions—consumers, producers, workers, or others? Why?**

💡 **SAMPLE ANSWER**

All of them: consumers, producers (including farmers and small businesses), workers, and experts.

A democratic government is accountable to all people. A decision that helps only one group (e.g. very low prices for consumers) may hurt another (farmers lose income). Hearing everyone gives fair, balanced decisions.

Questions and activities

1 **An increase in income always leads to a rise in demand for goods. Defend or refute, giving reasons for the same.**

✔ ANSWER

Refute (not always).

- Usually, a rise in income lets people buy more or better-quality goods, so demand for many goods rises.
- But demand also depends on tastes, needs and diminishing utility. When income rises, people may buy *less* of cheaper goods and switch to better ones (e.g. less coarse cloth or cheap rice, more branded items).
- Demand for things a person does not like or no longer needs will not rise.

So income generally raises demand, but not for every good.

2 **If petrol prices double, what happens to: a. Demand for diesel cars b. Demand for electric cars c. Demand for car accessories d. Demand for public transport**

✔ ANSWER

Petrol and petrol cars are complementary goods, so running petrol cars becomes costly.

- **a. Diesel cars:** demand rises (substitute for petrol cars), if diesel prices do not rise too.

- **b. Electric cars:** demand rises (substitute; no petrol needed).
- **c. Car accessories:** demand may fall, as fewer people buy or use petrol cars (complementary goods).
- **d. Public transport:** demand rises, as people switch from private cars to cheaper buses and metro (substitute).

3 **A farmer installs drip irrigation that reduces water use by 40% and increases yield by 30%. How does this affect: a. His cost of production b. His willingness to supply at different prices c. The overall market supply if many farmers adopt this technology**

✔ ANSWER

- **a.** Cost of production per unit falls: less water and less manual labour, and more output from the same field.
- **b.** He is willing to supply more at every price, because each unit is cheaper to produce and more profitable.
- **c.** Market supply increases (the market supply curve shifts to the right). With more supply, prices of the crop may fall, benefiting consumers.

4 **During online festival sales, the prices of many products are very low. Use the concept of demand and supply to explain why sellers**

sell at such a low price. What happens to the equilibrium when the price is lowered? Does this benefit only consumers or sellers as well?

✔ ANSWER

- Sellers often have large stocks (excess supply) and want to clear old stock before new models arrive. Lower prices increase quantity demanded (Law of Demand), so they sell much more.
- At the lower price quantity demanded rises a lot; as sales pick up and stock clears, the market moves towards a new equilibrium with a larger quantity sold.
- **Both benefit:** consumers buy goods cheaper; sellers earn more total revenue from high sales volume, clear stock, save storage costs and gain new customers.

5 **Suppose the government sets a maximum sale price for an essential vaccine below the market-driven price. What is likely to happen? a. Surplus b. Shortage c. No effect d. Fall in demand**

✔ ANSWER

b. Shortage.

A price ceiling below equilibrium raises quantity demanded (more people can afford it) but lowers quantity supplied (producers earn less and may produce less). Demand > supply → shortage. The government may need to support production or distribution to avoid it.

6 **The government levies higher taxes on tobacco and alcohol to promote healthier choices. Can you find out other goods where price controls have been set in place? What are the reasons for the same?**

💡 **SAMPLE ANSWER**

Examples:

- **Essential medicines:** maximum price fixed so that everyone can afford them.
- **Sanitisers in COVID-19:** price capped at ₹100 for 200 ml to stop hoarding and overcharging.
- **Minimum wages:** price floor on labour so workers earn enough.
- **Bus fares:** fixed so that public transport stays affordable.

Reasons: protect consumers from overcharging, help low-income groups, protect workers and farmers, and stop unfair practices like hoarding.

7 **Can excessive government regulation hurt markets? Explain with suitable examples.**

✔ ANSWER

Yes.

- **Price distortions:** if the government fixes wheat at ₹20/kg when the market price is ₹30/kg, farmers earn less, produce less, and shortages follow.
- **Compliance burden:** a small restaurant needs many permissions (food safety, fire safety, pollution, local clearances); the time and cost discourage small entrepreneurs (hurts ease of doing business).
- **Less innovation:** with price controls, farmers won't invest in better seeds, irrigation or technology, so long-term productivity falls.

8 **In the table, prices of guava are given: ₹100/kg, ₹80/kg, ₹50/kg, ₹20/kg.**

a. Write how much guava you will buy at each price.

b. Ask three friends and fill in the table.

c. Make a graph for each one of you and one final graph for the total quantity.

💡 SAMPLE ANSWER

Price	You	Friend 1	Friend 2	Friend 3	Total
₹100/kg	0.5 kg	0 kg	0.5 kg	1 kg	2 kg

₹80/kg	1 kg	0.5 kg	1 kg	1 kg	3.5 kg
₹50/kg	1.5 kg	1 kg	1.5 kg	2 kg	6 kg
₹20/kg	2 kg	2 kg	2 kg	3 kg	9 kg

Graphs: price on the y-axis, quantity on the x-axis. Each person's points joined give a downward-sloping individual demand curve. The 'Total' column gives the market demand curve, which is flatter (sum of all individual demand).

9 **Visit the nearby vegetable market and find answers:**

a. Who decides the prices of different vegetables?

b. Sometimes the prices of a few vegetables are too high, and sometimes too low. Why?

c. The price of tomatoes is high in the morning and lower by the evening. Comment.

💡 SAMPLE ANSWER

- **a.** No single person decides. Prices are set by the interaction of buyers and sellers (demand and supply); sellers also look at the wholesale (mandi) price and nearby sellers' prices.

- **b.** Because supply and demand change: season (in-season vegetables are plentiful and cheap), weather (rain or drought cuts supply), transport, festivals, and hoarding or rumours.
- **c.** In the morning vegetables are fresh and many buyers come, so demand is high. By evening vegetables are less fresh and perishable; sellers want to sell everything before they spoil and buyers are fewer, so sellers lower prices.

10 **Categorise into substitute goods and complementary goods:**

a. Movie ticket and popcorn

b. Eraser and pencil

c. Laptop and computer

d. Air conditioner and cooler

e. Notebook and pen

f. Apple and banana

g. Mobile and earphones

✔ ANSWER

Substitute goods	Complementary goods
c. Laptop and computer	a. Movie ticket and popcorn
d. Air conditioner and cooler	b. Eraser and pencil
f. Apple and banana	e. Notebook and pen
	g. Mobile and earphones

Substitutes can replace each other; complements are used together.

11 **Fig. 9.8 shows demand curve DD' and supply curve SS'. a. What does point E represent? b. Equilibrium price and quantity at E? c. Points A (on DD') and B (on SS') on the upper dashed price line: what do they indicate, and what is the gap AB? d. Points F (on DD') and C (on SS') on the lower dashed price line: what do they indicate, and what is the gap CF? e. If the price stays at the lower dashed line, what could happen next in a free market?**

✔ ANSWER

- **a.** E is where demand and supply curves cross: **market equilibrium** (quantity demanded = quantity supplied).
- **b.** Equilibrium quantity = **30 kg**; equilibrium price = about **₹250** (halfway between ₹200 and ₹300 on the graph).
- **c.** At the higher price (₹300), A shows the quantity demanded (about 23 kg) and B the quantity supplied (about 37 kg). Supply > demand, so the gap AB is **excess supply (surplus)**.
- **d.** At the lower price (about ₹160–170), F shows the quantity demanded (about 43 kg) and C the quantity supplied (about 17 kg). Demand > supply, so the gap CF is **excess demand (shortage)**.

- **e.** Buyers compete for the limited goods, so the price rises. As price rises, supply increases and demand falls, until the market reaches equilibrium at E.

12 **Draw a market equilibrium graph using:**

Price (₹)	10	20	30	40	50
Q.D. (kg)	5	10	15	20	25
Q.S. (kg)	25	20	15	10	5

a. Plot the demand and supply curve.

b. Identify the equilibrium price and quantity.

c. Analyse what happens if the price is set at ₹20 or ₹40.

💡 **SAMPLE ANSWER**

- **a.** Price on the y-axis (₹10–₹50), quantity on the x-axis (5–25 kg). Plot the Q.D. points and join them; plot the Q.S. points and join them. The two lines cross at one point.
- **b.** Q.D. = Q.S. = 15 kg at ₹30. **Equilibrium price = ₹30, equilibrium quantity = 15 kg.**
- **c.** As printed: at ₹20, Q.D. 10 kg < Q.S. 20 kg → excess supply of 10 kg; at ₹40, Q.D. 20 kg > Q.S. 10 kg → excess demand of 10 kg.

Note: by the Law of Demand, demand should fall as price rises, so the two rows seem swapped in the book. Then at ₹20: demand 20 kg > supply 10 kg → **shortage of 10 kg**, price rises; at ₹40: supply 20 kg > demand 10 kg → **surplus of 10 kg**, price falls. Either way the market moves back to ₹30 and 15 kg.

🎉 **Chapter 9 done! Revise the answers once more tomorrow.**